

WASP-12b

R-0353

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_240214 · created 2026-07-18T10:42:11+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460355.7081 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.011
Transit depth (Rp/Rs)^2	1.54 +/- 0.27 [%]
Orbital Inclination (inc)	79.82 +/- 2.18
Airmass coefficient 1 (a1)	1.0351 +/- 0.0099
Airmass coefficient 2 (a2)	-0.028 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#2 - [413, 210]
Optimal Aperture	4.6
Optimal Annulus	6.21
Transit Duration (day)	0.116 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.124 $\pm$ 0.011 vs 0.1178 (deviation 0.56 σ)
Duration fit vs published	0.116 d vs 0.125 d (deviation 1.23 σ)
Mid-time O-C	1567.9 min
Depth SNR	11.3
Residual scatter	0.95 %

Light curve

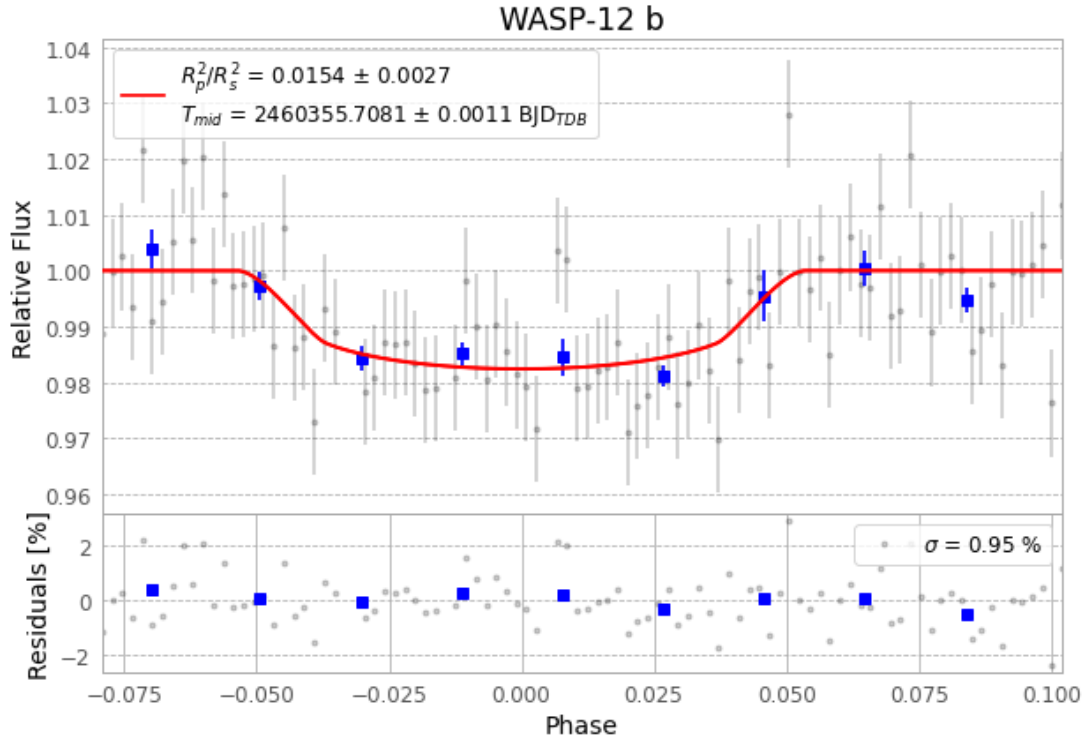


Figure 1 — FinalLightCurve_WASP-12 b_2024-02-14.png (SHA-256 f8207d76b33b411d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5952ab52fb93e685...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

96 FITS frames (63 MB), WASP-12240215024816.FITS → WASP-12240215073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240214.log (51c166e1da95...), FinalLightCurve_WASP-12 b_2024-02-14.png (f8207d76b33b...), FinalLightCurve_WASP-12 b_2024-02-14.csv (714fb2671165...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 10:42Z.